TELECOMMUNICATION ENGINEERING CENTRE
MOBILE TECHNOLOGIES DIVISION
STUDY PERIOD 2022-2024

NWG13-Report-6

NATIONAL WORKING GROUP 13

Original: English

Question(s): All/13 Virtual, 07th February 2024

MEETING REPORT

Source: Chairman, NWG13

Title: Meeting report of NWG13 (e-meeting, 07th February 2024)

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Abstract: This document contains the report for NWG13 e-meeting, on 07th February 2024.

CONTENTS

1. National Working Group 13 – corresponding to ITU-T SG13: Future networks and emerging network technologies

The NWG13 considered input contributions during its virtual meeting held on 07th February 2024 under the chairmanship of Shri Piyush Chetiya, DDG (MT), TEC. The group adopted the agenda in **NWG13-Agenda-5-R1**.

The objectives for this meeting were:

- To review the progress of on-going work items in ITU-T Y.ML-IMT2020-MLFO: "Architectural framework for MLFO in future networks including IMT-2020;
- To discuss others (based on contribution received - Request to TSB to conduct an analysis for the (i) proposal for establishing of the new SG13 Regional Group for Indian Ocean Rim countries (ii) creation of a Focus Group on Artificial Intelligence Native for Future Networks (FG AIFN) (iii) Draft Proposal for a new work item "Requirements for integrating Quantum Key Distribution in Internet Protocol Security (IPsec) to provide a secure Virtual Private Network" and (iv) advancement in the baseline text of
    - (a) draft new Recommendation ITU-T Y.ML-IMT2020-MLFO: "Architectural framework for MLFO in future networks including IMT-2020"
    - (b) Draft new Recommendation ITU-T Y.QKD-TLS: "Quantum Key Distribution integration with Transport Layer Security 1.3"
    - (c) Draft new Supplement ITU-T Y.MBIMT2020-Gen: "Information for Migrating Existing Network Technologies (2G, 3G, 4G) to IMT 2020 and beyond"
    - (d) Draft new Supplement ITU-T Y.supp.TC-QN: "Technical considerations towards Quantum Network"

NWG13 discussed 7 contributions. At the meeting, NWG13 produced 8 output documents including this meeting report.

The main results of the meeting are the following:

a) Proposal for establishing of the new SG13 Regional Group for Indian Ocean Rim countries

b) Contribution for creation of a Focus Group on Artificial Intelligence Native for Future Networks (FG AIFN)

c) Creation of a new work item "Requirements for integrating Quantum Key Distribution in Internet Protocol Security (IPsec) to provide a secure Virtual Private Network"

d) The updated text for the draft new Recommendation ITU-T Y.ML-IMT2020-MLFO: "Architectural framework for MLFO in future networks including IMT-2020"

e) Proposal for advancement in the baseline text of Draft new Recommendation ITU-T Y.QKD-TLS: "Quantum Key Distribution integration with Transport Layer Security 1.3"

f) Draft Proposal for advancement in the baseline text of Draft new Supplement ITU-T Y.MBIMT2020-Gen: "Information for Migrating Existing Network Technologies (2G, 3G, 4G) to IMT 2020 and beyond"

g) Draft Proposal for advancement in the baseline text of Draft new Supplement ITU-T Y.supp.TC-QN: "Technical considerations towards Quantum Network"

h) NWG13 meeting report (**NWG13-Report-06**).

2. Results

2.1 Contributions for Approval

Four contributions were presented by respective contributors for consideration of NWG13, so that the same could be submitted to ITU-T SG13 after approval of Competent Authority within the prescribed deadline of 20th February 2024for deliberation in 4th – 15th March 2024 meeting at Geneva.

2.2 Other documents for Approval

No Appendices, Supplements, Implementers' Guides or Technical Papers were proposed by NWG13 for approval at this meeting.

3. Outgoing liaison statements

No outgoing liaison statements were proposed by NWG13 for approval at this meeting.

4. Discussions

4.1. Proposal for establishing of the new SG13 Regional Group for Indian Ocean Rim countries

C-XX1	Vishnu Ram O.V./ Piyush Chetiya/Sujit Kumar/ Ziaur Rahman	The proposal for establishing of the new SG13 Regional Group for Indian Ocean Rim countries	Q All/13

- This document provides the rationale for the creation of a new Regional Group under SG13 designated for Indian Ocean Rim countries.

Meeting result

- The meeting agreed to accept the proposed text in the contribution with minor editorial changes.

4.2. Contribution for creation of a Focus Group on Artificial Intelligence Native for Future Networks (FG AIFN)

C-XX2	Vishnu Ram O.V./ Piyush Chetiya/Sujit Kumar/ Ziaur Rahman	Contribution for creation of a Focus Group on Artificial Intelligence Native for Future Networks (FG AIFN)	Q All/13

- This contribution proposes updated text for the draft Recommendation ITU-T Y.Trust-Registry: "Trust Registry for Devices and Applications: requirements, architectural framework". International Telecommunication Union (ITU) has approved the IMT-2030 (6G) vision framework. The usage scenarios identified by ITU for IMT-2030 (6G) needs an analysis of AI Native techniques which will enable these scenarios, especially the non-radio aspects. This document, is proposing to create a focus group to study the integration of AI Native techniques in future networks, building upon the previous work of ITU such as FG ML5G and FG AN. The new focus group will study the network impacts of integration of AI Native in future networks, the relevant AI Native techniques themselves and the validation mechanisms for such techniques.

Meeting result

- The meeting agreed to accept the proposed text in the contribution with minor editorial changes.

4.3. Requirements for integrating Quantum Key Distribution in Internet Protocol Security (IPsec) to provide a secure Virtual Private Network"

C-XX3	Rakesh Goyal/ Arjun Singh/ Mudita Chandra	Proposal for a new work item "Requirements for integrating Quantum Key Distribution in Internet Protocol Security (IPsec) to provide a secure Virtual Private Network"	Q 16/13

- This contribution proposes to start a new work item on "Requirements for integrating Quantum Key

Distribution in Internet Protocol Security (IPsec) to provide a secure Virtual Private Network”

Meeting result

- The meeting agreed to accept the proposed text in the contribution with some changes highlighted by Shri A S Verma and Ms. Rashmi Kamran.

4.4 ITU-T Y.ML-IMT2020-MLFO

C-XX4	Abhay Shanker Verma / Vishnu Ram O.V	Draft new Recommendation ITU-T Y.ML-IMT2020-MLFO: “Architectural framework for MLFO in future networks including IMT-2020”	Q 20/13

- This contribution proposes updated text for the draft new Recommendation Y.ML-IMT2020-MLFO “Architectural framework for MLFO in future networks including IMT-2020”.

Meeting result

- The meeting agreed to accept the proposed contribution.

4.5 Draft new Recommendation ITU-T Y.QKD-TLS: “Quantum Key Distribution integration with Transport Layer Security 1.3”

C-XX5	Rakesh Goyal	Proposal for advancement in the baseline text of Draft new Recommendation ITU-T Y.QKD-TLS: “Quantum Key Distribution integration with Transport Layer Security 1.3”	Q 16/13

- This contribution proposes advancement in the baseline text of Draft new Supplement ITU-T Y.QKD-TLS: “Quantum Key Distribution integration with Transport Layer Security 1.3”. The contribution proposes for amendment and addition of text in section 6 and 7

Meeting result

- The meeting agreed to accept the proposed contribution.

4.6 Draft new Supplement ITU-T Y.MBIMT2020-Gen: “Information for Migrating Existing Network Technologies (2G, 3G, 4G) to IMT 2020 and beyond”

C-XX6	Rakesh Goyal	Proposal for advancement in the baseline text of Draft new Supplement ITU-T Y.MBIMT2020-Gen: “Information for Migrating Existing Network Technologies (2G, 3G, 4G) to IMT 2020 and beyond”	Q 5/13

- This contribution proposes advancement in the baseline text of Draft new Supplement ITU-T Y.MBIMT2020-Gen: “Information for Migrating Existing Network Technologies (2G, 3G, 4G) to IMT 2020 and beyond” (Geneva, 23 October – 3 November 2023). The contribution proposes to include explanatory text added in section 7.2 and new text added in section 7.3.

Meeting result

- The meeting agreed to accept the proposed contribution.

4.7 Draft new Supplement ITU-T Y.supp.TC-QN: “Technical considerations towards Quantum Network”

C-XX7	Rakesh Goyal	Proposal for advancement in the baseline text of Draft new Supplement ITU-T Y.supp.TC-QN: “Technical considerations towards Quantum Network” including IMT-2020”	Q 16/13

- This contribution proposes advancement in the baseline text of Draft new Supplement ITU-T Y.supp.TC-QN: “Technical considerations towards Quantum Network” (Geneva, 23 October – 3 November 2023). The contribution proposes refinement in the baseline text of initial draft (Annex B).

Meeting result

− The meeting agreed to accept the proposed contribution.

4.8. Incoming liaisons, and others

None.

5. Closure

The NWG13 Chairman thanked the members for their participation in the NWG13 activities to progress the work, editors and contributors for their support and active involvement during this meeting.

Annex A:
List of the participants

No	Name	Affiliation
1	Abhijan Bhattacharyya	TCS Research
2	A S Verma	DOT
3	Ashish Vyas	DoT
4	Brajesh Kumar	TEC, New Delhi
5	DCPW	DCPW
6	Dileep Lakhera	Bharti Airtel Limited
7	Dr. Rashmi Kamran	IIT Bombay
8	Garima Mishra	TCS
9	Jyoti Roat (ADG)	TEC
10	M P Singh	NIT Patna
11	Piyush Chetiya, DDG(MT),TEC	TEC
12	Prof. Prabhat Kumar	NIT Patna
13	Rakesh Goyal TEC	TEC
14	Sujit Kumar	TEC
15	V K Roy,TEC	TEC
16	Vinod Kumar	TEC
17	Vishnu	Independent Expert
18	Ziaur Rahman	TEC

Annex B:
Summary of Q16/13 meeting activities

Date	Time	Agenda Items
07-February-2024 (Wednesday)	**16:00 hrs. onwards**	− Introduction − Draft proposal for establishing of the new SG13 Regional Group for Indian Ocean Rim countries. − Draft contribution for creation of a Focus Group on Artificial Intelligence Native for Future Networks (FG AIFN) − Creation of a new work item "Requirements for integrating Quantum Key Distribution in Internet Protocol Security (IPsec) to provide a secure Virtual Private Network" − The updated text for the draft new Recommendation ITU-T Y.ML-IMT2020-MLFO: "Architectural framework for MLFO in future networks including IMT-2020" − Proposal for advancement in the baseline text of Draft new Recommendation ITU-T Y.QKD-TLS: "Quantum Key Distribution integration with Transport Layer Security 1.3" − Draft Proposal for advancement in the baseline text of Draft new Supplement ITU-T Y.MBIMT2020-Gen: "Information for Migrating Existing Network Technologies (2G, 3G, 4G) to IMT 2020 and beyond" − Draft Proposal for advancement in the baseline text of Draft new Supplement ITU-T Y.supp.TC-QN: "Technical considerations towards Quantum Network"
Remote participation – https://cdotmeet.cdot.in/vmeet/a-s-9ld-wok		

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